Chapter test

24 Non-infectious diseases and disease prevention

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following statements about non-infectious diseases is correct?

   A. Non-infectious diseases are caused by pathogens.
   B. Non-infectious diseases can be transmitted from person to person.
   C. Non-infectious diseases are inheritable.
   D. Non-infectious diseases may be caused by stressful lifestyle. ☐

2. Which of the following descriptions about tumours is correct?

   (1) Benign tumours are localized.
   (2) Malignant tumours obtain nutrients by developing blood vessels towards body's main blood vessels.
   (3) Tumour cells lost their control in cell death.

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

★ 3. Which of the following diseases may be caused by smoking?

   (1) cardiovascular diseases
   (2) mouth and throat cancer
   (3) diabetes mellitus

   A. (2) only
   B. (1) and (2) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

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4. Which of the following components of cigarette will increase the heart rate of the smokers?

A. nicotine
B. tar
C. carbon monoxide
D. caffeine ☐

★ 5. Alcohol abuse would most likely cause damage to the

A. heart.
B. lungs.
C. liver.
D. blood vessels. ☐

6. Which of the following statements about cancer is ***incorrect***?

(1) Cancer arises when some cells divide too rapidly and become a malignant tumour.
(2) A diet high in fat but low in dietary fibre is a possible risk factor for liver cancer.
(3) Cancer can be treated by sulpha drugs.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

7. Which of the following is ***not*** a risk factor for cardiovascular diseases?

A. smoking
B. lack of exercise
C. over-exposure to UV light
D. old age ☐

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★ 8. Which of the following comparisons between insulin-dependent and non-insulin-dependent diabetes is correct?

	Insulin-dependent diabetes	***Non-insulin-dependent diabetes***
A.	usually occurs early in life	usually occurs later in life
B.	blood insulin level remains normal	blood insulin level is lower than normal
C.	target cells respond to insulin to a lesser extent	target cells respond to insulin normally
D.	insulin-producing cells in the pancreas grow excessively	insulin-producing cells in the pancreas are destroyed ☐

★ 9. Which of the following diseases can be prevented by vaccination?

A. hepatitis B
B. lung cancer
C. obesity
D. diabetes ☐

★ 10. Which of the following measures ***cannot*** prevent coronary heart diseases?

A. do not smoke
B. avoid exposure to ionizing radiation
C. have a balanced diet
D. have enough sleep and rest ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	24.1	24.1	24.1	24.1	24.1	24.1	24.1	24.1	24.2	24.2

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B. Problem solving (23 marks)

★ 11. Alcohol abuse is a social problem. Some countries have started to promote 'responsible drinking'. The recommended daily maximum alcohol intake for a responsible male drinker and female drinker are 4 units of alcohol and 3 units of alcohol respectively. 500 cm^3 of beer contains 2 units of alcohol.

(a) Suggest ***one*** possible reason for a female drinker to have a lower daily maximum alcohol intake. (1 mark)

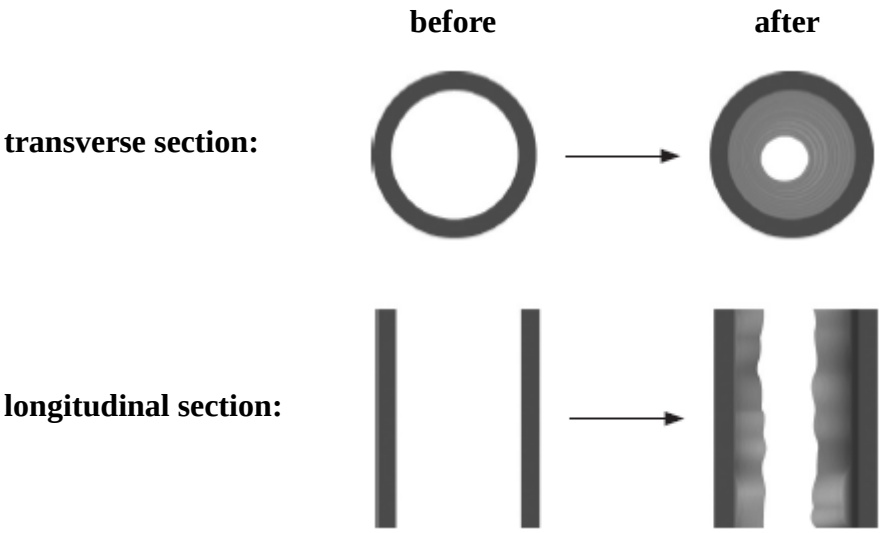
(b) State ***two*** long-term effects of alcohol abuse on the body. (2 marks)

(c) Explain why people with obesity are ***not*** advised to drink alcohol. (3 marks)

(Total: 6 marks)

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12. Coronary heart disease and stroke are the commonest cardiovascular diseases in Hong Kong. One of the main causes of cardiovascular diseases is the build-up of plaque in arteries. The diagrams below show the transverse sections and longitudinal sections of an artery before and after its walls are deposited with plaque.



(a) State ***two*** possible effects of plaque building up on an arterial wall. (2 marks)

(b) Coronary heart disease arises when plaque builds up in an artery which supplies blood to the cardiac muscle. Name this blood vessel and explain the effect of coronary heart disease on the cardiac muscle. (1 mark)

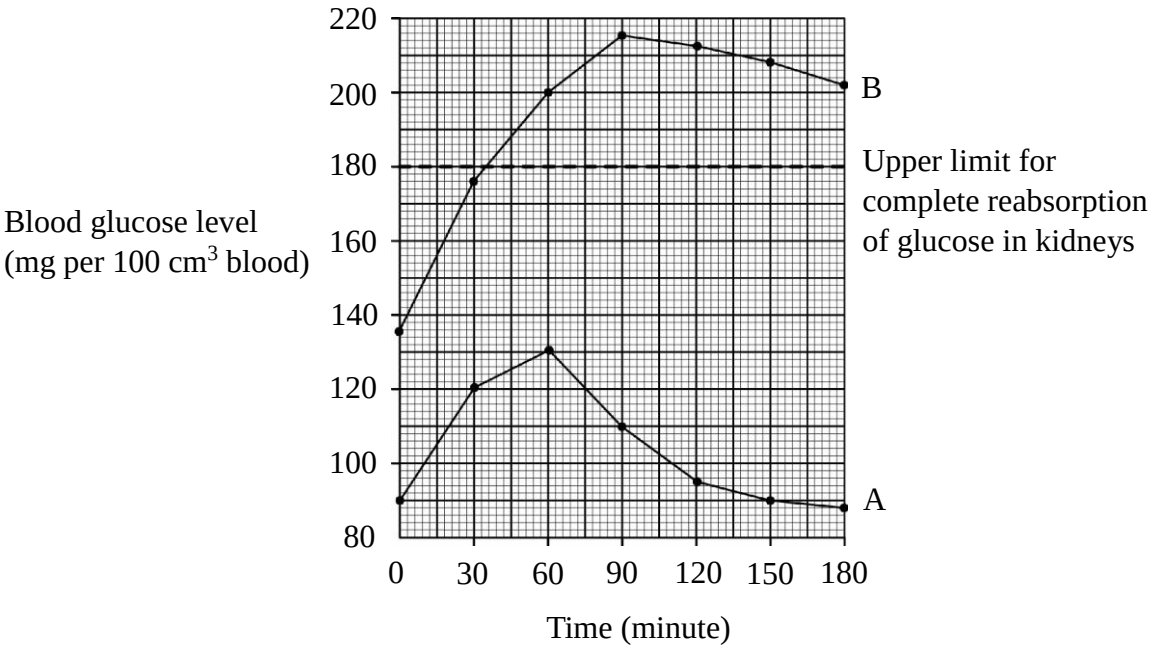
(c) In view of your answer to (b), suggest how to treat coronary heart disease. (2 marks)

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(d) Give ***two*** risk factors that would increase the incidence of stroke. (2 marks)

(Total: 9 marks)

★ 13. In a medical test, two persons A and B were fasted for 12 hours. They then drank equal volumes of glucose solutions of the same concentration. Their blood glucose levels were measured at 30-minute intervals for three hours. The graph below shows the changes in their blood glucose levels over three hours.



(a) Deduce and explain which person is a diabetic patient. (3 marks)

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The above medical test cannot differentiate the type of diabetes that the patient has. The doctor carried out another medical test by repeating the above test and measured the blood insulin level of the diabetic patient accordingly.

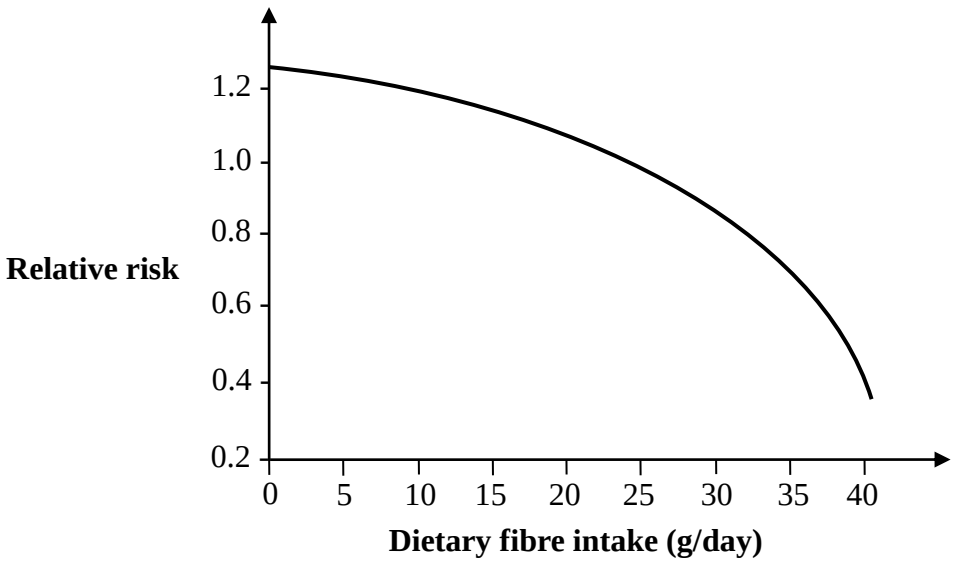
(b) Which type of diabetes does the patient have if the blood insulin level remains more or less the same? Explain your answer. (3 marks)

(c) State ***one*** cause for the type of diabetes you answered in (b) and suggest ***one*** possible treatment for this type of diabetes. (2 marks)

(Total: 8 marks)

C. Exam challenge (7 marks)

★★ 14. Studies show that unhealthy diet affects the risk of developing colorectal cancer. The graph below shows the relative risk of colorectal cancer in different groups of people who consume different amounts of dietary fibre daily.



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(a) Based on the graph, describe the relationship between daily dietary fibre intake and the relative risk of colorectal cancer. (1 mark)

(b) Based on your biological knowledge, suggest an explanation for the relationship you described in (a). (3 marks)

(c) Explain why the different groups of people being studied in the above investigation should be of the same age. (3 marks)

(Total: 7 marks)

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Marking scheme

A. Concept checking (10 marks)

1.	D	2.	D	3.	B	4.	A	5.	C
6.	C	7.	C	8.	A	9.	A	10.	B

(1 mark each)

B. Problem solving (23 marks)

11. (a) Women on average have a lower body weight than men. (1)

(b) Any ***two*** of the following: (2)

- liver damage / cirrhosis (1)
- mouth / pharynx / oesophagus / pancreas / liver cancer (1)
- coronary heart disease (1)
- high blood pressure (1)
- brain damage (1)

(c) Alcohol is produced from the fermentation of sugar-rich substances (e.g. barley or grape). (1)

Not all the sugars in the fermentation mixture are converted to ethanol, hence the resultant mixture contains alcohol and large amounts of sugars. (1)

Excess sugars will be converted into fat by the body. This would worsen the condition of obesity. (1)

(Total: 6 marks)

12. (a) Any ***two*** of the following: (2)

- the blood flow through the artery is reduced (1)
- the arterial wall becomes thickened / hardened (1)
- the artery becomes less elastic (1)
- the artery is blocked (1)

(b) coronary artery (1)

When the coronary artery is blocked, the oxygen and nutrient supply to the cardiac muscles becomes inadequate. (1)

The cardiac muscles around the blocked artery may die. (1)

(c) A vein or artery from other part of the body is cut and graft (1)

to by-pass the blocked region of the coronary artery. (1)

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(d) Any ***two*** of the following: (2)

- old age (1)
- high blood pressure / hypertension (1)
- a diet high in saturated fats and cholesterol (1)
- stressful lifestyle (1)
- smoking (1)
- alcohol abuse (1)
- lack of exercise (1)

(Total: 9 marks)

13. (a) Person B (1)
The blood glucose level of person B exceeded the upper limit for complete reabsorption of glucose in kidneys (1)
which would lead to excretion / presence of glucose in the urine. (1)

(b) Insulin-dependent diabetes / Type I diabetes (1)
This type of diabetes is caused by destruction of β cells in the pancreas which fails to secrete more insulin (1)
even when the blood glucose level rises above the normal level. The blood insulin level thus remains more or less the same. (1)

(c) Cause: hereditary (1)
Possible treatment: regular injection of insulin (1)

(Total: 8 marks)

C. Exam challenge (7 marks)

14. (a) The higher the daily dietary fibre intake, the lower the relative risk of colorectal cancer. (1)

(b) Dietary fibre stimulates peristalsis of the intestines, (1)
so food stays for a shorter period of time in the intestines. (1)
The lower the daily dietary fibre intake, the longer the time any carcinogens in food may stay in and affect the intestines. (1)

(c) The risk of cancer increases with age. (1)
Older people are at higher risk because they have been exposed to carcinogens for a longer time / their immune systems are weaker and are less effective in removing cancer cells. (1)
Studying people of the same age can eliminate the influence of ageing on the risk of cancer development. (1)

(Total: 7 marks)

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